

KAGGLE · BUILD WITH AI — GEMMA HACKATHON · GDG WINDSOR

GEMMA MONSTERS

A monster-battling citadel with a real tutoring agent inside.

Five monsters are out to get the student — each with its own favourite way of tripping them into a mistake.

Grade 9 EQAO mathematics. Gemma through Ollama, entirely on one laptop.

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A wrong answer is not noise. It is a sentence.

$$\frac{2}{3} + \frac{1}{4} = \frac{3}{7}$$

Tops with tops, bottoms with bottoms. Feels tidy. Completely wrong.

$$\frac{11}{12}$$

the answer
rescaled both
tops to twelfths

$$\frac{3}{7}$$

added straight across
never looked for
a denominator

$$\frac{3}{12}$$

found the denominator
then forgot to
rescale the tops

one question · three different heads · one right idea and two different wrong ones

A wrong answer is a specific wrong idea that feels right.

These two students do not need the same worksheet. They need different ideas found

Five monsters, five favourite ways to trip you

Each monster is out to get the student, and each has its own preferred trap. Below, in the student's own words, is what it wants them to believe.

Fractis

Number · 17 items

“add the tops, add the bottoms” · “multiplying powers? multiply the exponents”

Equazor

Algebra · 13 items

“the sign stays put when it crosses the equals sign” · “distribute to the first term, then stop”

Statiq

Data · 5 items

“when in doubt, take the mean” · “the mode has to be the biggest number”

Polygor

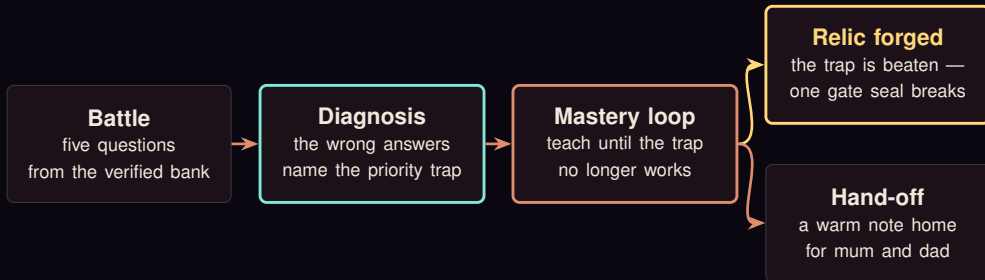
Geometry & Measurement · 9 items

“double the side, double the area” · “radius, diameter — near enough”

Ledgerling

“simple or compound, interest is interest” · “the smaller

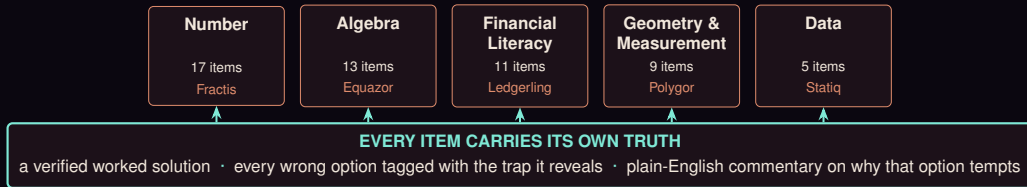
One session, start to finish



- The monster meets the student **by name**, and it **remembers**: Gemma writes its greeting from the real battle history on disk.
- Between battles: **ninety-second mental-maths skirmishes** against the Collector's three lieutenants.
- Beat all five monsters → the golden gate opens → **the rescue finale**.

The bank is the ground truth

Ontario **MTH1W** — the course behind the Grade 9 EQAO assessment.
55 items, audited against the published expectations and the EQAO framework.



The answer key is balanced 14 / 14 / 14 / 13 across A–D, which removes the single-letter guessing edge — always picking C scores no better than chance. The wrong-answer commentary is keyed to the **answer text**, never the option letter, so it stays true however the options are ordered. Retrieval is a table lookup, not a search: the payload wanted is an exact answer key.

One student, one trap: the battle question

1 THE BATTLE QUESTION - BANK ITEM DAT-Q2

"What is the median of this data set? 14, 3, 20, 8, 11"

The student picks **11.2**. The key says **11**.

2 THE TRAP, NAMED BY TABLE LOOKUP — NO MODEL CALL

Mean by Default — the tag on that option, with the bank's own note: *11.2 is the mean, $56 \div 5$.*

3 GROUND TRUTH, CARRIED INTO THE PROMPT

Order first: 3, 8, 11, 14, 20. The middle of five is 11.

Gemma teaches rung 1, **Direct correction**, and may state no other number.

The question, the tag, the note and the solution all come from the verified bank. Gemma is handed the answer; it is asked only to explain why the other route felt right.

The same trap, checked on fresh ground

4 CHECK - A FRESH BANK ITEM, SAME TRAP

"0 pets: 7 students. 1 pet: 9. 2 pets: 5. 3 pets: 4. What is the **mode** number of pets?"

Option D is 1.24 — the mean again, in a new coat.

5 THE STUDENT ANSWERS 1, AND WRITES IN THE REASONING BOX

"The mode is the value with the tallest bar, not an average at all."

The box is optional. This student filled it, so Gemma grades those words: **RE-SOLVED**.

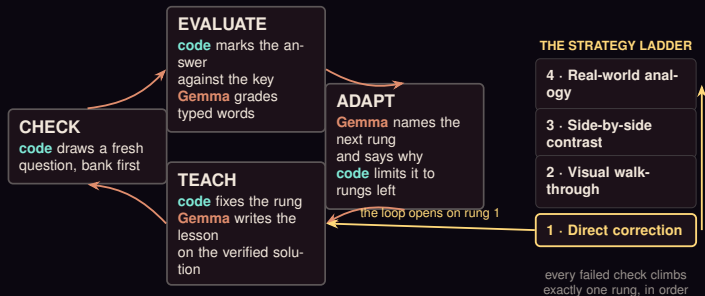
6 THE MASTERY DECISION - PLAIN CODE

One fresh question correct, with reasoning that held up. One more and Statiq falls.

A thin explanation would not have advanced the streak; one still carrying the trap resets it.

Correctness is decided by the bank's key. Only the words the student typed are ever handed to the model for judgment.

The loop: teach until the trap stops working



Coral = Gemma judges. Teal = code guarantees. Most steps are both, so each node names its two halves.

Deterministic spine. Plain code runs the loop and owns every transition; Gemma is called only at the judgment points.

The bar: two fresh questions correct in a row. When the student explains their thinking, that explanation is graded, and a thin one does not advance the streak.

Three independent stops: 4 check attempts, 12 model calls, a spent ladder. Any one ends the session, so it cannot loop forever.

Every change of strategy is announced on screen, with its

When the student explains, the explanation is graded

The reasoning box is optional — leaving it blank never blocks anyone. Fill it in, and those words decide whether the streak moves.

THE CHECK QUESTION

Calculate $\frac{1}{2} + \frac{1}{3}$ — picks $\frac{5}{6}$ (correct)

THE STUDENT TYPES

"I just guessed the biggest one"

THE GRADE

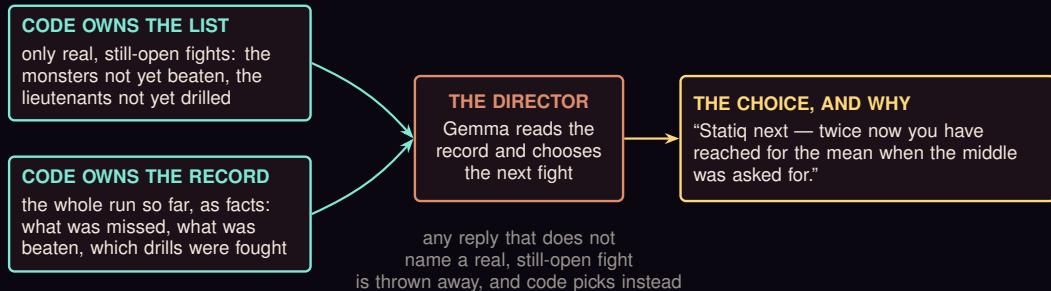
SHALLOW — the streak does **not** advance

THE CITADEL ANSWERS BACK

"Luck is a coin, challenger. The monsters ahead do not take coins."

- Three labels, nothing else — **RESOLVED** / **SHALLOW** / **SAME_ERROR**. A closed set the model chooses from, never open-ended prose.
- **SAME_ERROR** on a *right* answer resets the streak: correct answer, but the thinking underneath it walked into the same trap.
- The citadel replies **to the student's own words**, in character — effort met seriously, a joke met with a warning.
- **It fails open**. Any parse problem returns **RESOLVED**, so a model hiccup can never cost the student anything.

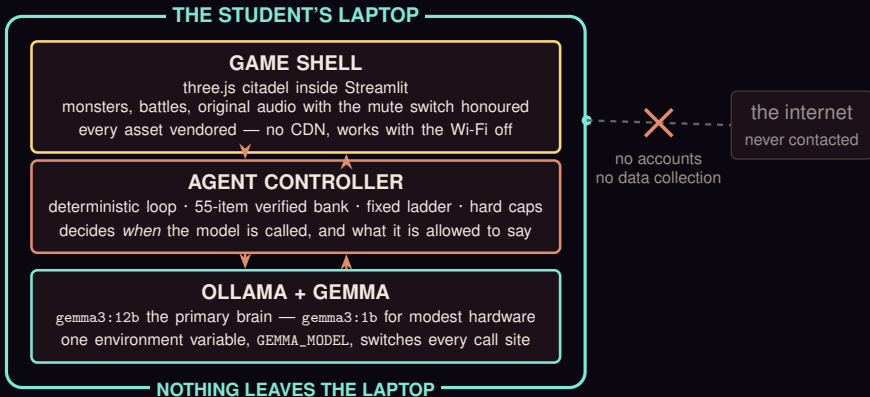
The agent decides where the student goes next



Everywhere else Gemma answers one question. Here it **shapes the session** — and it must name the evidence it used.

The record it sees is complete, so it cannot invent a history the student never had.

Everything on one laptop



Every asset is vendored into `static/`. The only network call the code makes is to **Ollama on localhost**, and the letters home are written to a file on the same machine.

Gemma judges. Code guarantees.

WHAT GEMMA DECIDES

THE JUDGMENT CALLS

writes every lesson and study guide
grades the student's typed reasoning
picks the next rung, and says why
authors fresh questions, under audit
voices the monsters and the citadel
phrases the note for mum and dad

WHAT CODE GUARANTEES

THE GROUND TRUTH

every mathematical fact, from the bank
answers marked against the real key
the trap named by tag lookup
the ladder order, fixed in advance
the caps that end every session
every number a parent ever reads

THE GROUNDING RULE, ON THE CURRICULUM PATH — wherever a bank item is being taught or checked, its verified worked solution rides inside the prompt:

Gemma explains *why a method fails* and may state no number the bank did not supply. Two calls sit outside that path by design — the mental-maths strategy line before a speed drill and the coach's after-drill feedback — where there is no bank item to ground against and no answer is being marked.

Every mathematical fact a student is graded on comes from the bank — **never from the**

Designed for the model to fail

Four things the model actually did in testing — and the structure that now stands in the way.

It invented arithmetic mid-explanation

The grounding rule. On the curriculum path the verified solution rides in the prompt and is the only maths the model may state. It explains *why* a method fails; it never re-does the sum.

It marked its own key wrong

The blind self-solve audit. Bank first, always. A Gemma-authored question reaches a student only if Gemma can solve it blind and agree with its own key.

It fabricated counts in prose

Code prints every number; the model prints none. It wrote “beaten three traps” when one was beaten — a figure borrowed from another count. Any digit now discards the reply.

It answered in Spanish

English is written into the prompts that ask Gemma to author new text — the three fresh-question writers and the citadel's reply. Small models drift; those prompts do not leave it to chance.

The part that reaches the kitchen table

LETTERS HOME

FOR MUM AND DAD

A note goes home after **any battle with mistakes**, on hand-off, and on wins.

Each names the trap in plain words and ends with **three things to try at the kitchen table**.

They persist. Notes are written to disk, so closing the tab does not lose them, and a pinned button reaches them from every screen.

A PRINTABLE SHEET

ONE TRAP, ONE PAGE, NO SCREEN

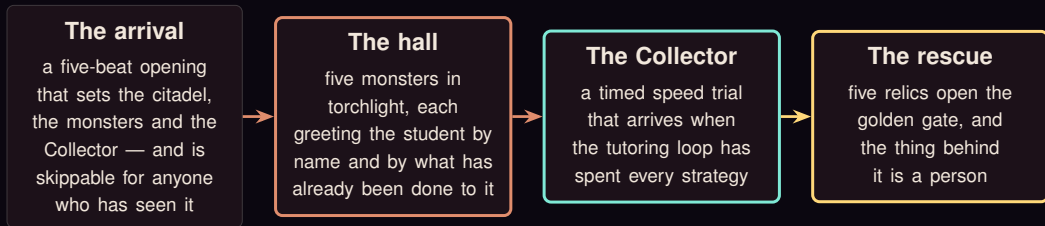
Up to **ten questions on the one trap** the student is stuck on, with room to write and an **answer key on its own page**.

Bank first, always. Anything generated passes the **same blind self-solve audit**.

A wrong key costs one retry on screen.
On paper it would cost a parent's evening.

Every number a parent reads is **computed in code**. Gemma only says what they mean.

And a fourteen-year-old wants to play it



- The Collector keeps **three lieutenants** — Twinfang for doubling and halving, The Niner for nines, Splitjaw for making tens. Ninety seconds each, and a Gemma coach that reads **the actual misses** rather than the score.
- Original audio throughout, and the **mute switch is honoured everywhere**, including the places nobody checks.

**Find the idea behind the wrong
answer.**

**Teach until the student can
show it is gone.**

Bank-grounded, hard-capped, and entirely on one laptop.

A student who adds fractions straight across meets Fractis, is taught on a fixed ladder of four strategies, and the loop ends one of two ways — two fresh questions correct in a row, or a note written home. Nothing about them ever leaves their laptop.